

Electricity demand: combining recent and weekly forecast-error memory

P100-041 · Source-aware scientific reference. Supported as a scoped predictive extension; novelty and demonstrated industrial impact are not established.

Scenario and measured endpoint

Unimputed hourly balancing-authority demand in MW. Retrospective published-snapshot forecast correction. Input official forecast for target hour, preceding-hour forecast, and actual/forecast residuals 48 and 168 hours before target. Nominal issuance $t-24$ h is an analysis convention, not observed release timestamp. Load-history timestamps precede nominal issuance by at least 24 h. Forecast vintages and actual public release times are unknown; this is not an as-issued operational test. No target-hour demand, generation or interchange predictor.

Authority-month groups; rolling train Jan–Feb->March, Jan–March->April, Jan–April->May. June confirmation. Same authorities across time, correlated national weather; no independence claim across all authorities. Development contains 181,474 eligible observations in 265 groups. Confirmation contains 37,889 eligible observations in 53 groups. Unimputed native Demand/Forecast only. Revised final public snapshot is not an as-issued operational backtest. Negative/nonfinite demand/forecasts excluded by validity rule before fitting. Primary MW MAE is not normalized by authority load.

Findings and executable relation

Complementary recent and weekly forecast errors provide a compact bias-transport correction. The mechanism is persistent forecasting/operating context, not electrical power-balance identity. Retrospective error reduction does not demonstrate dispatch cost, reliability or causal industrial impact.

The reference selected before confirmation is attempt-008:

$$\widehat{D}_t = F_t + a(D_{t-48h} - F_{t-48h}) + b(D_{t-168h} - F_{t-168h})$$

D and F are native unimputed demand and official demand forecast (MW). a and b are dimensionless robust development-fitted weights. The two historical residuals are at least 24 h older than the nominal $t-24$ h issuance convention. Forecast snapshots may contain revisions; exact original publication timestamps are unavailable.

Coefficient	Exact fitted value
a	0.4974490504629177
b	0.3525440289648312

All numerical constants, transformations, alternative equations and fitted coefficients are executable in run.py and serialized in rules.json. Any algebraic consequence of this equation is an implication of the same fitted model, not a second independently validated finding.

Experimental method and evidence

10 substantive development attempts tested competing representations, controls and ablations. Fits use equal-group weighted least squares or declared robust loss, with all coefficients and radial-basis centers learned only inside each training fold. Selection minimizes mean group MAE, preferring fewer coefficients within 1% of the minimum. Separate confirmation checks the frozen states and performs no fitting. Baselines share permitted information.

Primary errors are in MW; each group contributes equally regardless of row count.

Frozen model	Development MAE	Confirmation MAE	Worst confirmation group
reference	215.716	283.872	2274.46
flexible	344.508	569.108	4742.24
official	337.849	464.341	4293.15
residual48	247.569	323.862	2813.04
weekly	627.461	1005.93	11294.9
attempt-001	254.819	331.817	2929.84
attempt-002	227.967	296.154	2457.33
attempt-003	331.767	452.411	4176.67
attempt-004	232.272	301.764	2452.72
attempt-005	240.255	306.017	2457.33
attempt-006	215.836	283.972	2273.22
attempt-007	227.501	295.69	2449.59
attempt-008	215.716	283.872	2274.46
attempt-009	235.551	310.679	2722.07
attempt-010	244.212	329.406	2627.81

The frozen reference changes confirmation MAE by 12.3% relative to the strongest evaluated baseline `residual48` (323.862MW); it wins on 50/53 whole groups. This is descriptive, not a population significance test. Rows within runs, days, weeks, profiles or casts are dependent. The complete group errors, bias, RMSE and 90th-percentile errors are in `evidence/by_group.csv`.

Falsification, limits and value

- Multiplicative-response fitting is not interchangeable with physical-MW error optimization. - The development-selected correction is a retrospective forecasting extension, not a new conservation law or an as-issued operational trial. - Report whole-group errors and ranges; no independent-row population CI. Temporal/spatial dependence and small number of sites/floats limit inference. - Public data and known physical parameterizations preclude automatic novelty certification. Predictive accuracy does not establish an industrial intervention or a new physical law.

Use the existing official forecast as a strong baseline and respect actual publication delays. Physical-unit validation exposes failures that relative-error fitting can conceal. Retain a simpler model when diurnal terms add no meaningful evidence.

Reproduction and future evaluation

Run `python run.py` in this folder to verify allowlisted hashes and reproduce every saved confirmation prediction. This executes trusted bundled reference code; scoring an external prediction file should use the shared Principia evaluator without executing submitted code. Numerical alternatives need not resemble this equation. Use the exact registered task, timing/input budget, units, eligibility and group weighting. New endpoints or information access require a separately reviewed task version.

All confirmation outcomes are now exposed to future users. Source-stage exposure is recorded in `task_spec.json`; the first-use internal freeze history is preserved in research history. Independent source data are needed for a new confirmation campaign.

Primary sources and existing analyses

- <https://www.eia.gov/opendata/browser/electricity/rto/region-sub-ba-data> — Official EIA dashboard describes hourly actual/day-ahead forecast demand and interchange. Forecast error correction is an established statistical operation; the source offers no as-issued revision archive in this local bundle. - https://www.eia.gov/outlooks/aeo/nems/documentation/electricity/pdf/EMM_AEO2025.pdf — EIA-930 collects balancing-authority operating data. Final snapshots must not be represented as immutable real-time information availability.